

Some Techniques For LOWERING POWER CONSUMPTION



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While very-large-scale integration (VLSI) and complementary metal oxide semiconductor (CMOS) technologies have had an exponential growth, energy-storage technologies have grown slowly, requiring improvement in electronic circuit designs. Today, such ICs are embedded in all kinds of electronic devices. These devices work on direct current (DC), which is often derived from alternating current (AC) mains supply. The conversion process using an AC adaptor, or a battery itself, consumes significant amount of power.

High power consumption areas

Power consumption is often directly related to performance. Higher the power available for consumption, the better can be the performance of the device. Hrishikesh Kamat, chief executive officer, Shalaka Technologies, says, "Optimising speeds and performance is a challenge." A wearable device performing powerful functions, for example, would result in high power consumption. Lowering power consumption would limit its functions.

Losses through heat are a significant part of power consumption. "For smaller devices, heat losses are negligible or sometimes almost non-existent," says T. Anand, managing director, Knewron Technologies. This becomes a problem in bigger devices where lot of power is lost through heat, requiring heat-sinks. Although higher speeds also cause higher power consumption, it is better to work faster in a short span of time. The device can go to sleep mode, whenever feasible, to reduce power consumption.

A step in this regard has been the

introduction of low-power microcontroller modules. "Cortex M4 architecture based controllers are coming with deep sleep and hibernation," says Kamat. "You can increase power consumption temporarily with an interrupt, and the system goes back to using lesser power."

CMOS circuits for low power

With increase in integration, size and complexity of chips, thermal management is adding significant cost to the electronic products. Dynamic power dissipation in ICs does not increase much beyond 65nm, while static or leakage power also more or less remains the same. Localised dynamic power consumption can be reduced by providing different frequency clocks to different blocks of an IC.

Threshold voltages in CMOS cells also govern IC's characteristics. Sub-threshold current in a CMOS IC reduces when threshold voltage is increased. Cells with a higher voltage can help in the reduction of sub-threshold current. Combining low- and high-voltage cells to reduce leakage current, while maintaining the desired frequency of operation, is a typical design trend.

High-voltage cells are used as 'sleep' transistors, which gate the supply to further low-voltage based designs when the block is in standby mode. Sleep transistors are turned on, resulting in low-voltage blocks downstream, during active mode.

Design trends, with a catch. By varying bias voltage using a control circuit, voltage can be dynamically changed as per the required application. It requires much complex MOS fabrication in twin-well or triple-well fabrication techniques, more commonly known as variable threshold CMOS (VTCMOS).

Lowering voltage, however, compromises with the reliability of the chip, causing voltage swings. These voltage swings can arise due to various process or environmental variations.

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